

READ.ME WORDEDITION

OBJECTIVE;

The objective of this project was to design and simulate a circuit component in series and also in parallel, using Ltspice in order to understand the relationship between voltage, current and the resistance of different component when connected together in series, parallel or series-parallel.

CIRCUIT COMPONENT

1. Dc voltage source
2. Wires
3. Resistors

THEORETICAL CALCULATION

Series component/connection;

This circuit have all component connected by a single wire with the same current flowing through every component; voltage is divided among the component.

Resistors in series;

$$R_t = R_1 + R_2 + R_3 + \dots + R_n$$

I, current remain the same although while voltage changes

Voltage can be calculated using ohms law; $V_t = I_t * R_t$

Individual voltage can be calculated using

$$V_1 = I_t * R_1$$

$$V_2 = I_t * R_2 + \dots$$

PARALLEL CONNECTION;

This voltage remain the same while the current is split among each components are connected across the same two nodes.

$$R_t = 1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n$$

V_t remain the same although the component

We can find current using ohms law; $V_t = I_t * R_t$

$$I_t = I_1 + I_2 + I_3 + \dots + I_n$$

SERIES-PARALLEL CIRCUIT

This is a combination of both the series circuit and the parallel circuit. it contain portions of the circuit connected in series and other portions connected in parallel.

Kirchhoff's current law(KCL)

The total current entering a node equals to sum of the individual current leaving the node. $\sum I_{in} = \sum I_{out}$

$I_{in} = I_{out}$

Where I_{in} is the total current entering into the node

I_{out} is the total current leaving the node

Kirchhoff's voltage law (KVL)

State that the algebraic sum of all the voltage changes around a closed loop is zero

$V_{in} - V_{out} = 0$ V_{in} is the total voltage entering the loop while V_{out} is the total voltage leaving the loop

Theory vs Simulation

From the calculation using Ohm's law, you can calculate the voltage drop across each resistor when the component is connected in series while the current will remain constant although the series connection. Voltage remains constant when going through resistor components laid out one after another while current changes as it flows through the component in parallel connection.

Simulation; from simulation it can be seen there is actually a voltage drop across each resistor when they are connected in series, same current flows through the circuit. While voltage remains the same when applied across resistors in parallel but the current changes as it moves through the circuit.

WHAT I HAVE LEARNED

From the simulation, I learned that:

1. Current remains the same at any point in a series connection
2. Current varies across each path in a parallel circuit
3. Voltage drop across each resistor in goes through in a series circuit
4. Voltage remains the same across each path in a parallel circuit
5. In a closed loop, the amount of voltage entering is equivalent to the amount of voltage leaving the loop. $V_{in} - V_{out} = 0$
6. The sum of all the current entering a node is equal to the current exiting the node

Engineering Applications

Some of the engineering applications include

1. Conservation of energy; energy supplied is equal to the energy being used
2. Conservation of charge; what goes in must come out
3. Resistors connected in parallel would have a lesser resultant resistance than the one connected in series